

Plugin A: "RetroCrusher" (Bitcrusher)

A destructive audio effect for the targeted reduction of digital audio resolution.

GUI Features

- **Knobs:** Rotary controls for Bit Depth and Downsample.

DSP Mathematics & Core Logic

Time-Crushing:

An internal `holdCounter` keeps the audio signal constant over a defined number of samples, thereby chopping the temporal resolution horizontally into chunky steps.

Volume-Crushing:

The amplitude values are scaled to a positive range (0.0 to 1.0), multiplied by the number of target bit steps, truncated using `std::floor`, and then scaled back.

Safety Features:

Integration of `buffer.clear()` for unused output channels to eliminate garbage data from the host computer's RAM.